Z-motion correction

validation & benchmarking

optimouse 2-photon pipeline — C57_O1M2/10022023

CaImAn 1.12.2 · Suite2p 0.14.0 · real 41-plane z-stack

1. End-to-end pipeline validation

Validation of the current method on real anatomy

The z-motion correction chain was validated on real data (C57_O1M2/10022023):

mcorr → z-motion (F_anat) → ROI extraction → per-ROI fit-and-subtract.

Coverage:

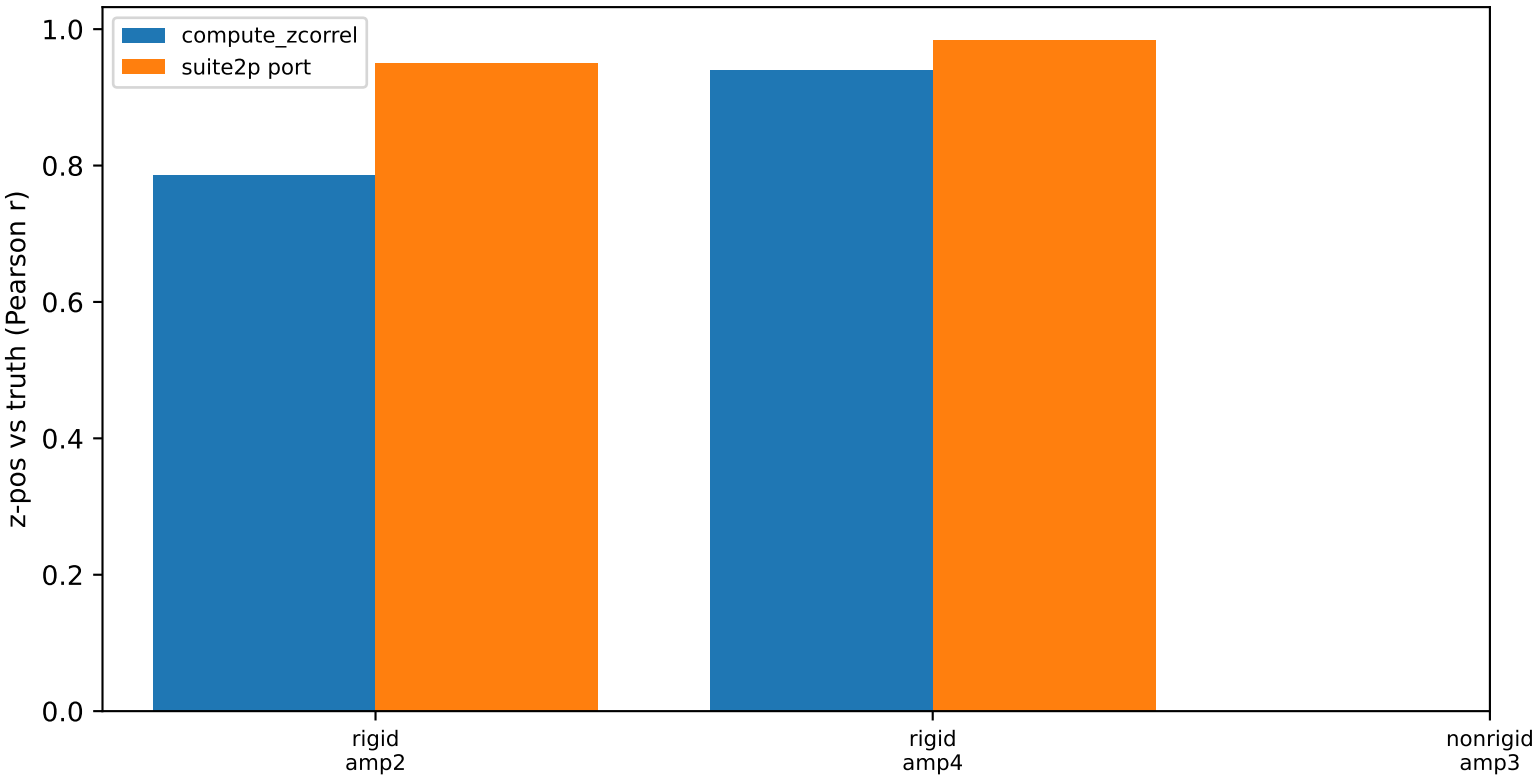
- Subset (TSeries-001, 200 frames) and full session (1600 frames).
- Both extraction backends: CNMF and Suite2p (0.14.0).
- Full session: real z-drift over planes 9–16 (~7 µm, σ≈1 µm); 593 ROIs → F_roi_zcorrected.

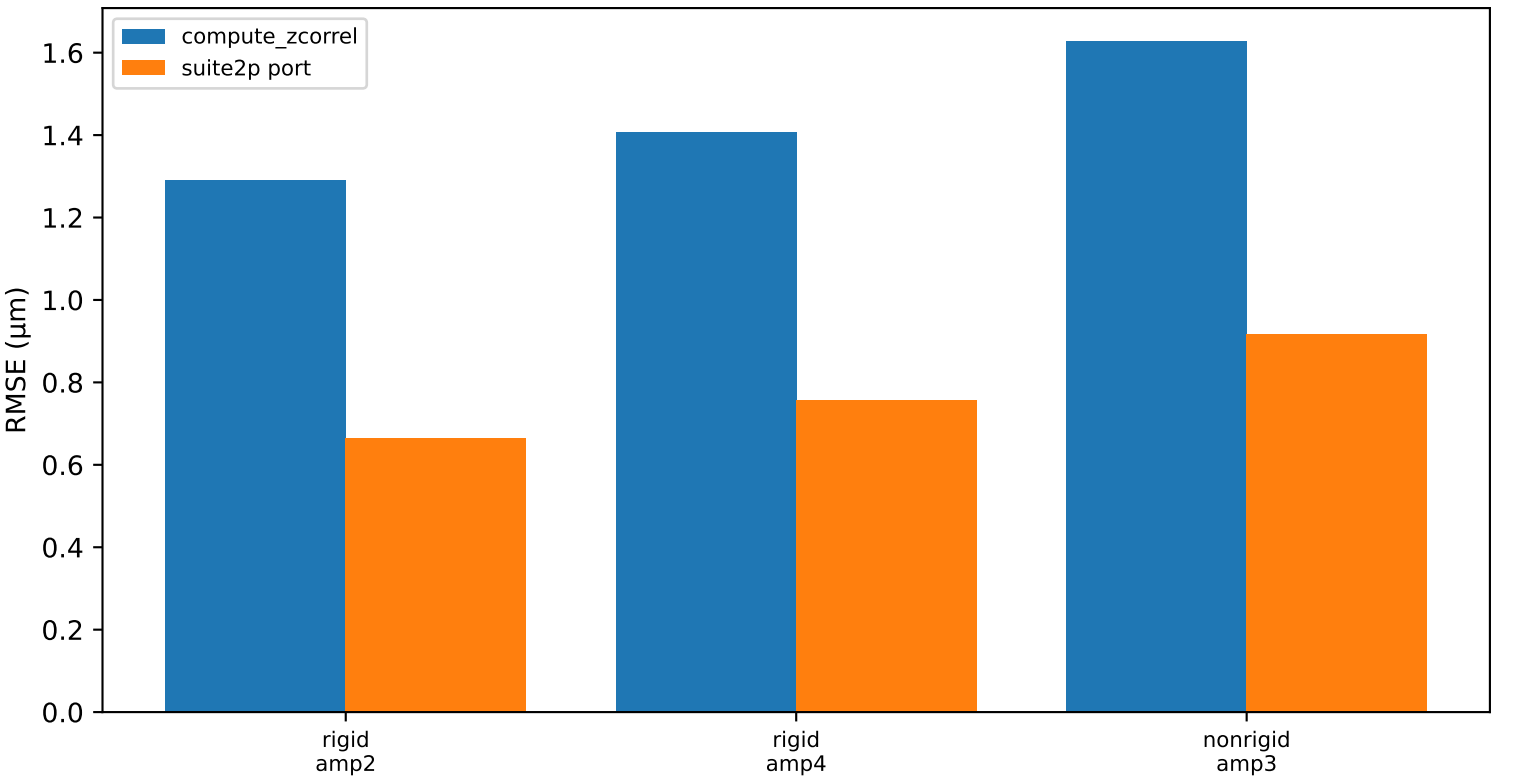
7 bugs fixed in the under-tested per-ROI path (normally done in MATLAB):

- create_components_movie excerpt overflow on short recordings.
- CNMF extraction now writes cnmf_result.hdf5 for run_roi_zcorr.
- load_CNMF(n_processes=1); raw F = C+YrA; dims from cnm.dims (caiman 1.12).
- compute_zcorrel_suite2p: contiguous flip + worker-scope rigid (suite2p 0.14).

Environment: pixi (CaImAn 1.12.2 + Suite2p 0.14.0 pinned to production). The pipeline requires an activated env — bare `python` may be Python 2.

2. Stage 1 — z-position method comparison (compute_zcorrel vs Suite2p port)





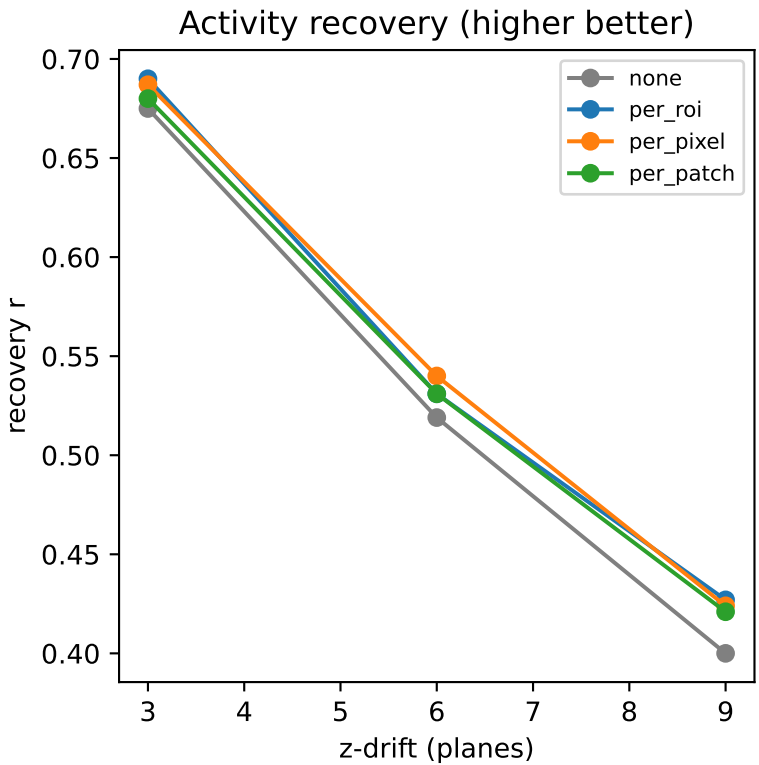
Suite2p is more accurate (higher r, lower RMSE) but slower — time/case: rigid amp2: 13.8s vs 82.6s rigid amp4: 2.6s vs 7.9s nonrigid amp3: 2.8s vs 8.0s

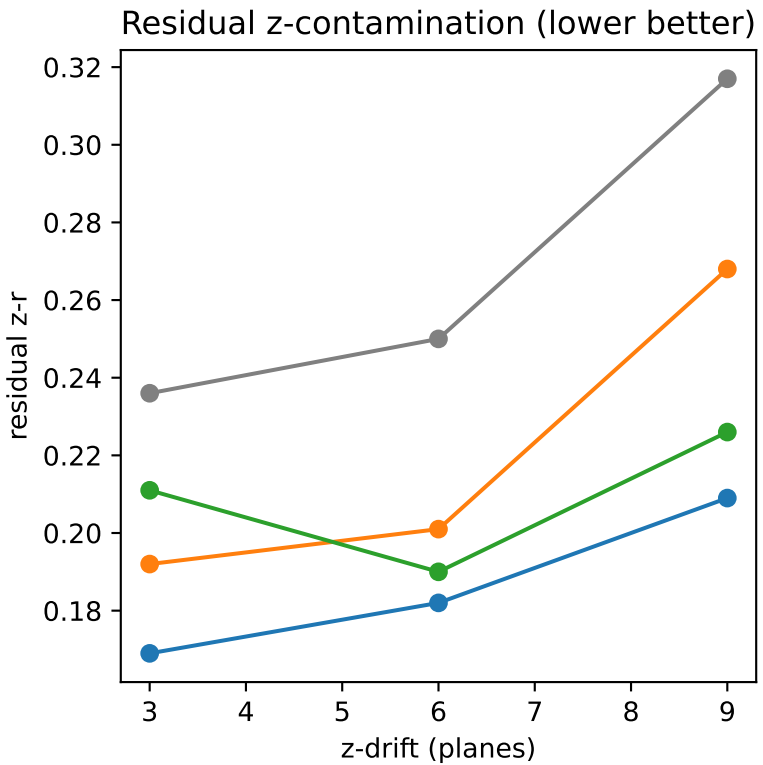
3. Stage 1b — sub-plane interpolation (tested)

case	metric	integer argmax	sub-plane
global z, amp 1	RMSE planes	2.258	2.261
global z, amp 2	RMSE planes	2.790	2.774
global z, amp 4	RMSE planes	1.808	1.791
F_anat corr, amp 3	recovery r	0.721	0.722
F_anat corr, amp 6	recovery r	0.548	0.549

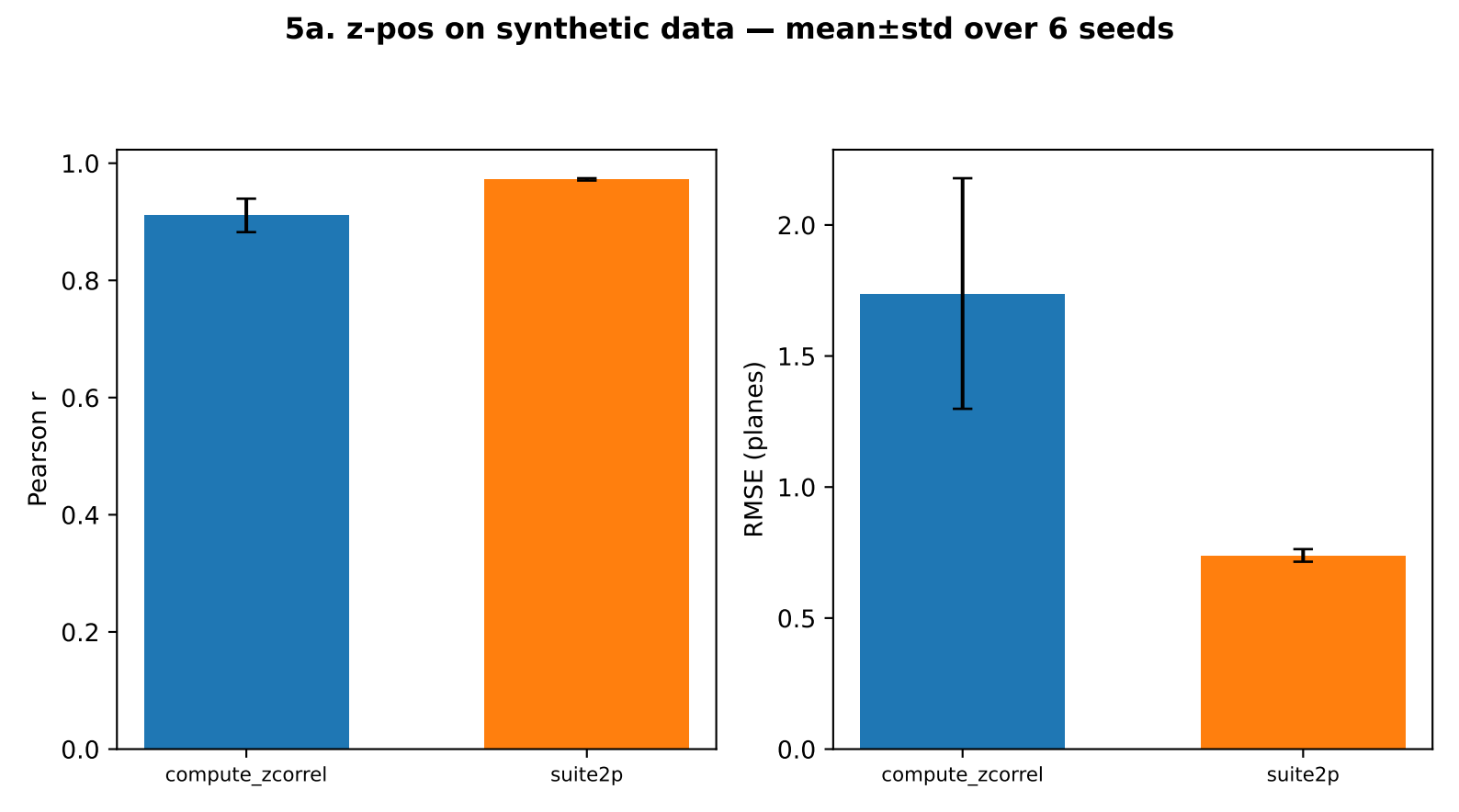
Parabolic interpolation of the correlation peak was expected to beat integer argmax for both the global z-position and the per-patch depth used to build F_anat. Tested: negligible improvement in both. The error is dominated by discrete multi-plane misassignments and defocus signal attenuation, not ±0.5-plane quantization. Conclusion: sub-plane is not worth adding; improve plane-selection robustness or use the more accurate Suite2p estimator instead.

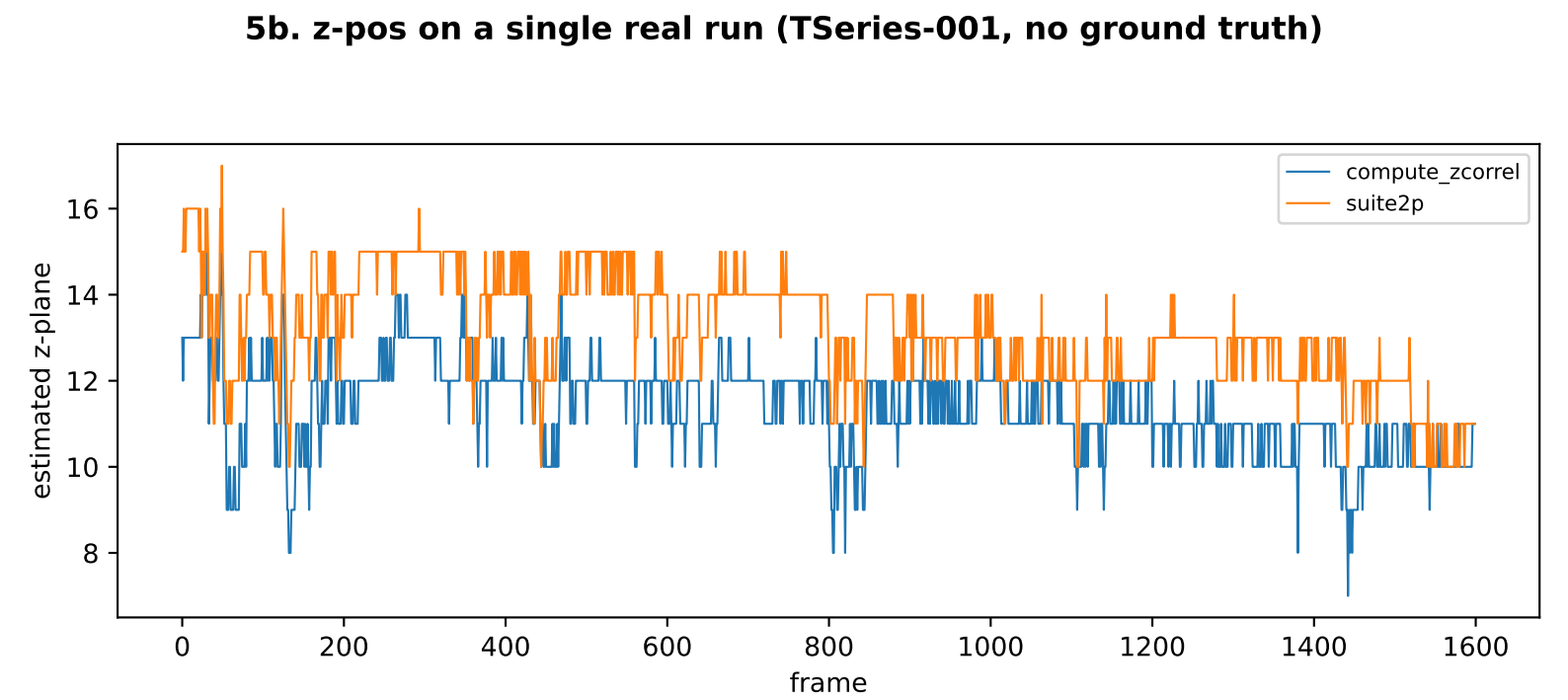
4. Stage 3 — correction method comparison (synthetic activity ground truth)

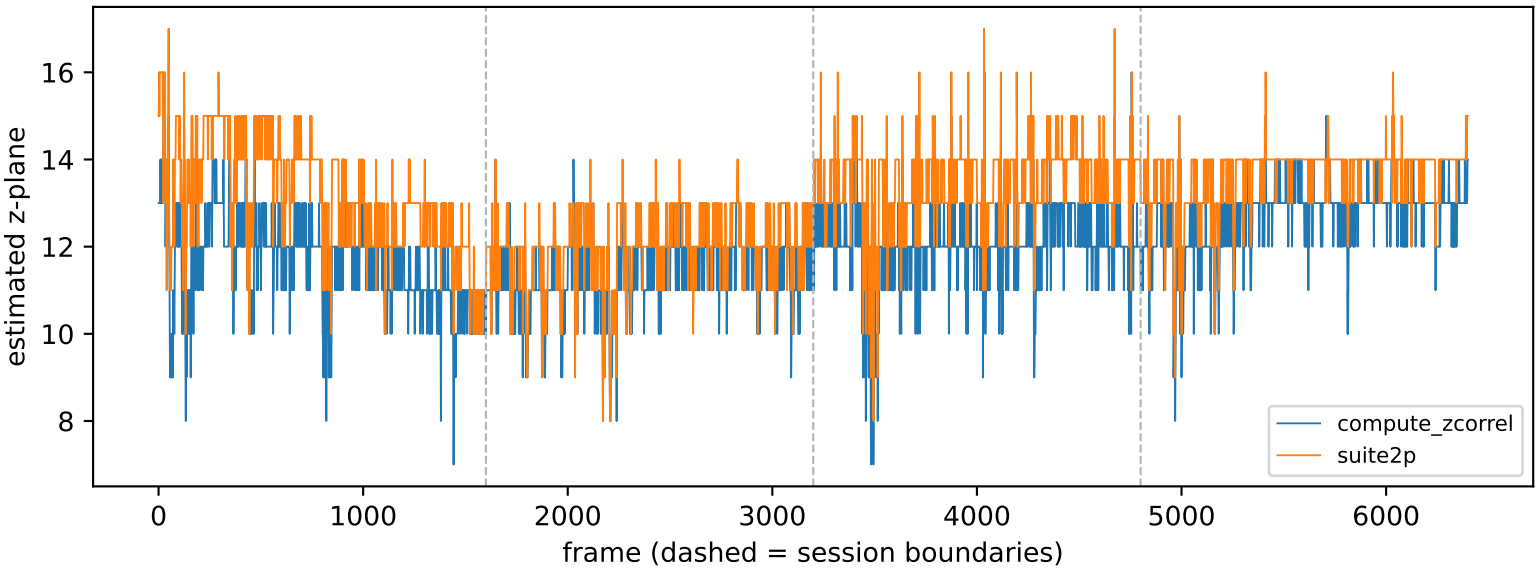




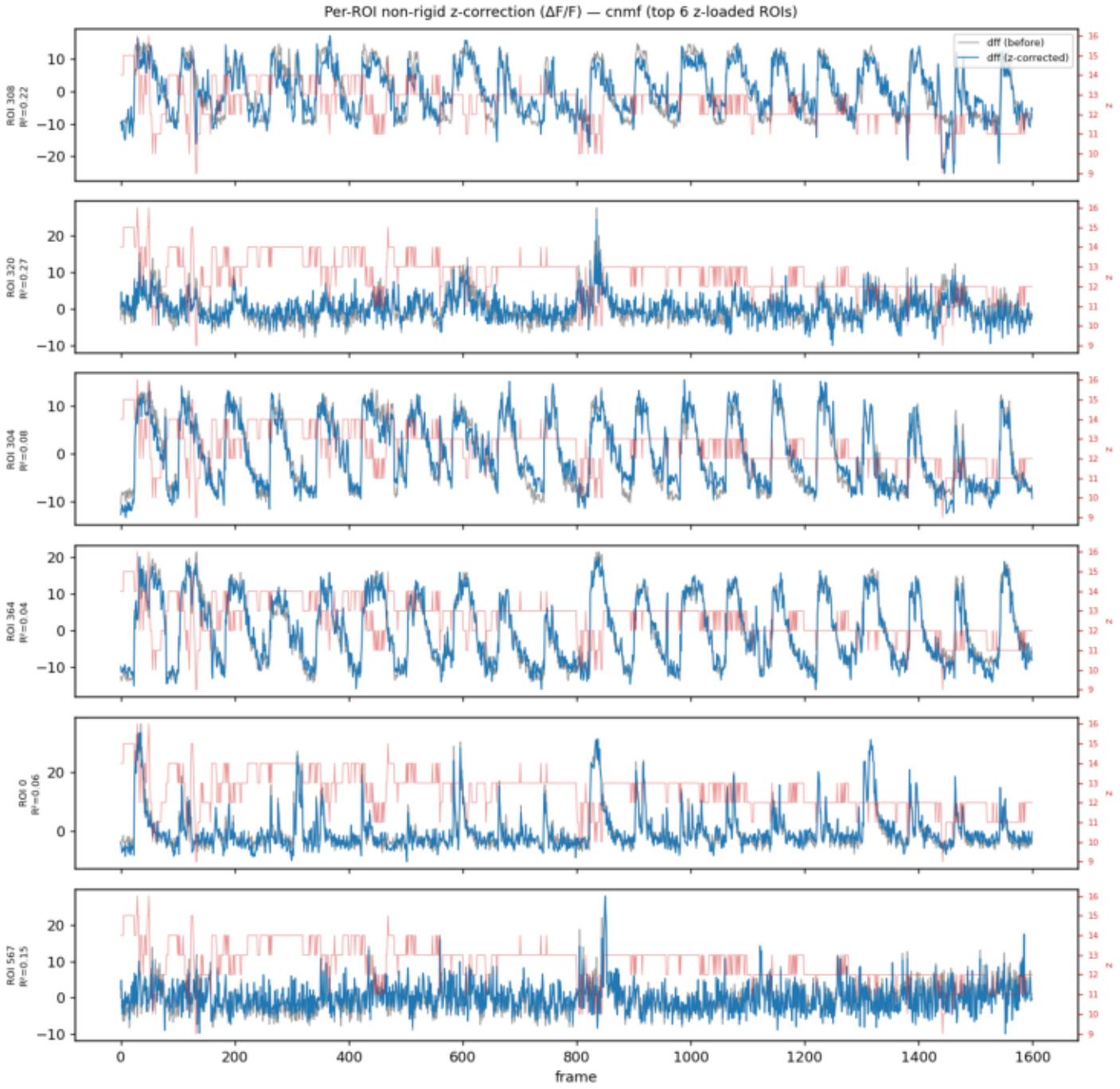
per-ROI removes the most z-contamination at every amplitude; recovery is capped by defocus attenuation.


5a. z-pos on synthetic data — mean±std over 6 seeds
Pearson r
1.0
0.8
0.6
0.4
0.2
0.0
compute_zcorrel
suite2p
RMSE (planes)
2.0
1.5
1.0
0.5
0.0
compute_zcorrel
suite2p


5b. z-pos on a single real run (TSeries-001, no ground truth)
compute_zcorrel
suite2p
estimated z-plane
16
14
12
10
8
0
200
400
600
800
1000
1200
1400
1600
frame
agreement r=0.748; ≥2-plane jumps: compute_zcorrel=24, suite2p=44 (fewer = smoother)

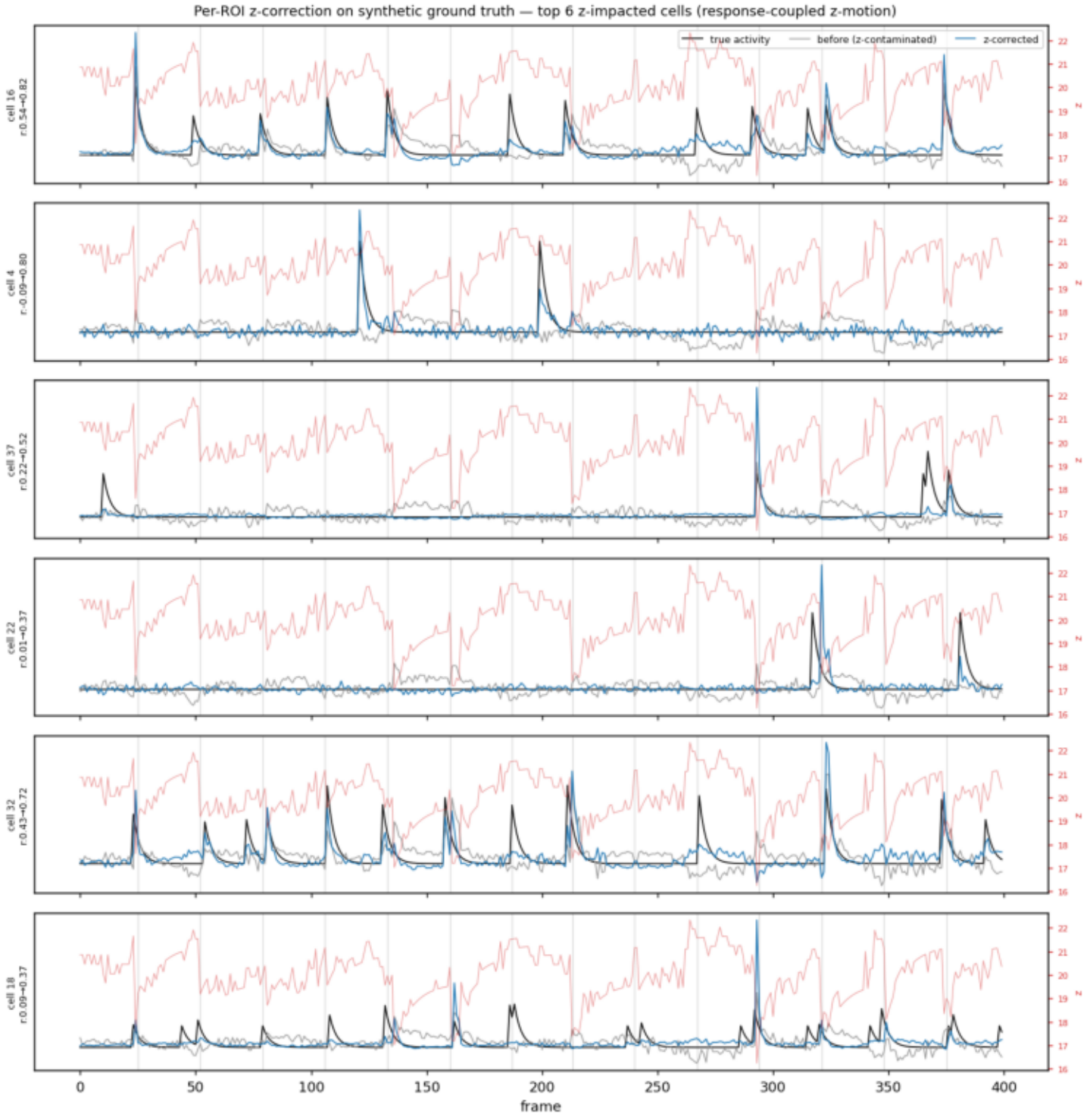

5c. z-pos on concatenated 4-session data
estimated z-plane
16
14
12
10
8
0
1000
2000
3000
4000
5000
6000
frame (dashed = session boundaries)
compute_zcorrel
suite2p
mean |z jump| at boundaries: compute_zcorrel=1.00, suite2p=0.33 planes

6. Real-data per-ROI correction (full TSeries-001)



6c. Correction on synthetic ground truth

On real data the benefit is hard to see (no ground truth). Injecting known activity into the real z-stack lets us quantify recovery. The z-motion is modelled on the real protocol: fast dynamics (from the real estimated zpos) plus downward motion onsets clustered around stimulus responses (grey vertical lines) — so the z-artifact partly overlaps the true responses, the hardest case. For the most z-impacted cells the correction removes the z-jitter (grey→blue) and recovers the true transients (black).



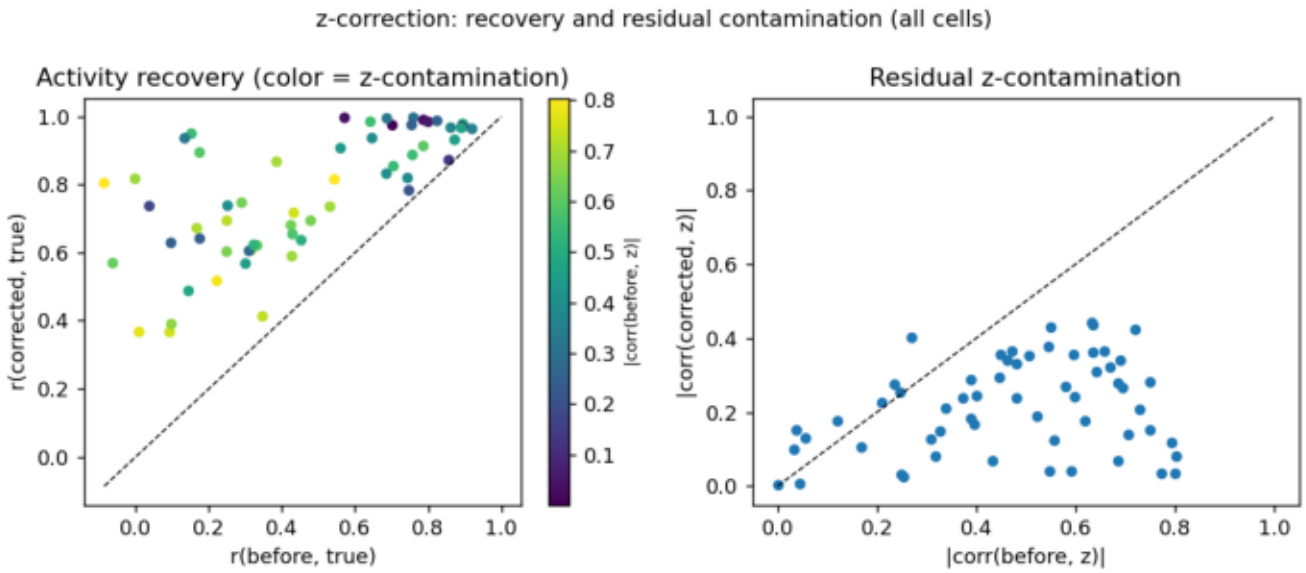
6c. Correction on synthetic ground truth — recovery

All cells: recovery r 0.47 → 0.78 | residual z-contamination 0.47 → 0.22

Top z-impacted (10): recovery r 0.24 → 0.62 | residual z 0.75 → 0.17

Benefit: the most z-impacted cells more than double their activity recovery and shed ~85% of z-contamination. Clean cells are left essentially unchanged.

Limit: recovery plateaus below 1.0 — z-defocus attenuates the true signal, which additive subtraction cannot restore.



6b. Optimized compute_zcorrel

compute_zcorrel_legacy = full-frame Pearson (kept for comparison).

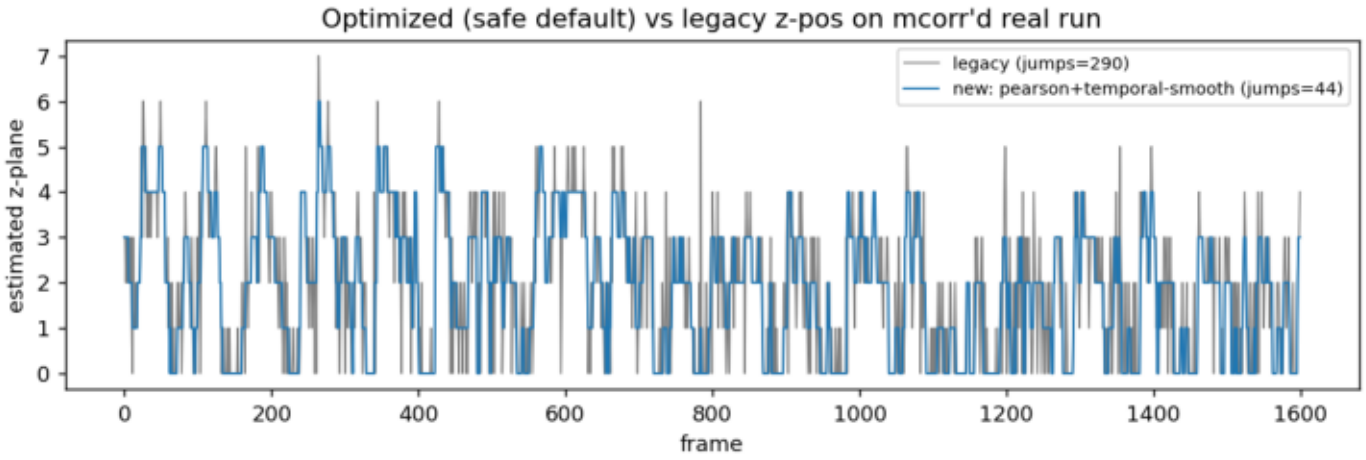
New compute_zcorrel default = Pearson + temporal median smoothing of zpos.

Synthetic ground truth (4 seeds): suite2p/phasecorr is most accurate (RMSE ~0.7 planes) but — see below — fails on mcorr'd data. Pearson+smoothing edges out legacy (r 0.935 vs 0.920).

Two synthetic-motivated options were tested and left OFF by default because they DEGRADE on NoRMCorre-corrected production movies:

- phasecorr (Suite2p): latches onto the deepest low-contrast plane (→ plane 40).
- edge_crop: collapses the Pearson estimate to plane 0.

The one safe win — temporal smoothing — cuts spurious ≥2-plane jumps on the real mcorr'd run from 290 (legacy) to 44, tracking the same drift (r=0.85):



7. Conclusions & recommendations

What works, and what to do next

Pipeline:

- The z-motion chain is validated end-to-end on real data, both CNMF and Suite2p backends. 7 latent bugs were fixed in the per-ROI path.

z-position estimation:

- Synthetic ground truth: phase correlation (Suite2p) is most accurate (RMSE ~0.7 vs ~1.7 planes for Pearson).

- BUT on real NoRMCorre-corrected movies this does not transfer: phasecorr latches onto the deepest plane, and edge-cropping collapses Pearson to plane 0. Since z-pos runs after mcorr, neither is usable there.

- The robust improvement is Pearson + temporal median smoothing of zpos: same drift as legacy but 44 vs 290 spurious ≥2-plane jumps on the real run.

- Recommendation: keep Pearson (now the default in compute_zcorrel) with temporal smoothing; do NOT adopt phasecorr/edge-crop for production despite their synthetic accuracy. Revisit phasecorr only for raw/lightly-registered data.

F_anat & correction:

- Per-ROI fit-and-subtract removes the most z-contamination at every drift amplitude; per-pixel/per-patch trail. Recovery is capped by defocus attenuation.

- Sub-plane interpolation of the z-peak gives negligible gain (global or F_anat) — not worth adding.